

HAKODATE, Japan

WMO: 474300

Lat: 41.8167N

Lon: 140.7542E

Elev: 44

StdP: 100.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-8.3	-14.0	1.1	-7.4	-12.7	1.3	-6.3	11.4	0.5	10.3	-0.7	3.0	310	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	28.4	23.2	27.1	22.5	25.9	21.8	24.2	27.2	23.5	26.2	22.7	25.2	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.1	25.9	22.5	17.3	25.4	21.7	16.5	24.8	73.2	27.4	70.4	26.1	67.4	25.3	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.3	8.2	DB	-12.0	30.4	1.8	1.6	-13.3	31.6	-14.3	32.5	-15.3	33.5	-16.6	34.6	(4)
			WB	-12.6	24.9	1.7	0.9	-13.8	25.6	-14.8	26.1	-15.8	26.6	-17.1	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-2.7	-2.0	1.7	7.3	12.4	16.4	20.7	22.4	18.9	12.5	5.8	-0.2	(6)
(7)		DBStd	9.18	3.00	3.26	3.20	2.99	2.71	2.55	2.69	2.49	2.84	3.19	3.97	3.46	(7)
(8)		HDD10.0	1552	395	336	258	91	9	0	0	0	13	133	317		(8)
(9)		HDD18.3	3490	653	569	516	331	184	69	9	2	26	181	376	574	(9)
(10)		CDD10.0	1366	0	0	0	10	83	191	331	384	268	91	8	0	(10)
(11)		CDD18.3	263	0	0	0	0	1	9	82	127	43	1	0	0	(11)
(12)		CDH23.3	1113	0	0	0	0	3	22	288	645	154	1	0	0	(12)
(13)		CDH26.7	158	0	0	0	0	0	1	32	108	17	0	0	0	(13)
(14)	Wind	WSAvg	3.6	3.7	3.8	4.2	4.0	3.5	3.2	3.0	3.1	3.3	3.6	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	1195	78	65	64	72	93	85	124	154	148	104	113	94	(15)
(16)		PrecMax	1580	199	202	112	134	158	182	334	362	275	197	221	173	(16)
(17)		PrecMin	882	42	23	17	7	21	28	27	24	33	34	53	45	(17)
(18)		PrecStd	160	33	38	24	34	33	47	64	79	61	45	44	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	8.6	12.6	19.2	23.4	25.6	29.2	30.6	28.6	22.2	17.6	11.5	(19)
(20)		MCWB	4.4	5.9	8.2	12.1	14.9	18.4	23.7	24.3	23.2	17.2	13.3	9.2	(20)	
(21)		2%	DB	4.7	6.1	10.4	16.4	20.9	23.6	27.5	29.1	26.5	20.6	15.6	9.1	(21)
(22)		MCWB	2.6	3.4	6.8	10.4	14.0	17.6	22.6	23.8	21.9	16.0	12.1	7.0	(22)	
(23)		5%	DB	3.3	4.6	8.7	14.3	19.1	22.1	26.1	27.8	25.0	19.3	14.0	7.0	(23)
(24)		MCWB	1.2	2.2	5.4	9.4	13.3	17.3	22.0	23.0	20.4	15.1	10.9	4.7	(24)	
(25)		10%	DB	2.1	3.3	7.1	12.6	17.5	20.8	24.9	26.6	23.8	18.1	12.5	5.3	(25)
(26)		MCWB	0.1	1.0	4.1	8.4	12.6	16.8	21.4	22.4	19.7	14.4	9.7	3.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	6.4	9.2	13.3	16.6	20.1	24.4	25.4	24.1	19.0	14.4	9.5	(27)
(28)		MCDWB	6.0	8.1	11.3	16.5	20.8	22.9	27.5	28.6	27.3	21.0	16.2	11.0	(28)	
(29)		2%	WB	2.9	3.9	7.5	11.7	15.3	19.0	23.6	24.6	23.0	17.2	12.9	7.2	(29)
(30)		MCDWB	4.4	5.7	10.0	14.8	19.0	21.8	26.3	27.7	25.5	19.3	15.0	9.0	(30)	
(31)		5%	WB	1.5	2.5	5.9	10.3	14.3	18.1	22.9	23.9	21.9	16.1	11.3	5.0	(31)
(32)		MCDWB	3.0	4.3	8.0	13.3	17.8	21.0	25.4	26.9	24.1	18.5	13.6	6.6	(32)	
(33)		10%	WB	0.3	1.2	4.5	9.1	13.3	17.3	22.0	23.2	20.5	15.0	9.9	3.2	(33)
(34)		MCDWB	1.9	3.1	6.8	12.1	16.6	20.1	24.4	25.9	22.9	17.7	12.1	5.1	(34)	
(35)	Mean Daily Temperature Range		MDBR	6.2	6.7	7.2	8.2	8.0	6.9	5.9	6.2	7.7	8.5	7.4	6.1	(35)
(36)		5% DB	MCDBR	6.9	7.7	9.2	10.6	11.1	9.1	7.5	7.7	8.4	9.0	9.6	8.0	(36)
(37)		MCWBR	6.0	6.3	6.7	6.5	6.0	4.8	3.7	3.9	4.8	5.8	7.5	7.0	(37)	
(38)		5% WB	MCDBR	6.9	7.7	8.6	9.3	9.2	7.5	6.3	6.6	7.0	7.9	9.2	7.9	(38)
(39)		MCWBR	6.5	6.9	7.2	6.5	5.6	4.4	3.4	3.7	4.6	5.9	7.9	7.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.355	0.401	0.473	0.529	0.546	0.518	0.514	0.482	0.424	0.407	0.392	0.371	(40)	
(41)		taud	2.116	1.991	1.880	1.806	1.853	2.008	2.092	2.199	2.316	2.221	2.154	2.101	(41)	
(42)		Ebn at Noon	753	772	759	749	750	773	771	785	808	765	706	690	(42)	
(43)		Edh at Noon	106	141	178	206	201	173	158	138	114	111	102	100	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.05	3.27	4.47	5.00	4.89	4.29	4.19	3.69	2.75	1.59	1.12	(44)	
(45)		RadStd	0.08	0.18	0.23	0.47	0.46	0.43	0.42	0.39	0.34	0.20	0.15	0.08	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	(46)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page